MSG No. 139.535 $I4'/mmm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4'/mmm'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32]`

Table 2: Wyckoff site: `2b`, site symmetry: `4'/mmm'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32]`

Table 3: Wyckoff site: `4c`, site symmetry: `mmm.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[9,10,11,12,13,14,15,16]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[17,18,19,20,21,22,23,24]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[25,26,27,28,29,30,31,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4'm2'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,4,6,7,27,28,29,30]`
2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[2,3,5,8,25,26,31,32]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[9,10,15,16,18,19,21,24]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[11,12,13,14,17,20,22,23]`

Table 5: Wyckoff site: `4e`, site symmetry: `4'mm'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4,6,7,9,10,15,16]`
2	$[0,\ 0,\ -z]$	`[2,3,5,8,11,12,13,14]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[17,20,22,23,25,26,31,32]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[18,19,21,24,27,28,29,30]`

Table 6: Wyckoff site: `8f`, site symmetry: `..2'/m'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,16,21,28]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[2,13,22,25]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[3,14,23,26]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[4,15,24,27]
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[5,12,17,32]
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[6,9,18,29]
7	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[7,10,19,30]
8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[8,11,20,31]

Table 7: Wyckoff site: 8g, site symmetry: 2mm.

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,4,6,7]
2	$\left[0, \frac{1}{2}, -z\right]$	[2,3,5,8]
3	$\left[\frac{1}{2}, 0, z\right]$	[9,10,15,16]
4	$\left[\frac{1}{2}, 0, -z\right]$	[11,12,13,14]
5	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[17,20,22,23]
6	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[18,19,21,24]
7	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[25,26,31,32]
8	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[27,28,29,30]

Table 8: Wyckoff site: 8h, site symmetry: m.2'm'

No.	position	mapping
1	$[x, x, 0]$	[1,8,11,16]
2	$[x, -x, 0]$	[2,7,10,13]
3	$[-x, x, 0]$	[3,6,9,14]
4	$[-x, -x, 0]$	[4,5,12,15]
5	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[17,24,27,32]
6	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	[18,23,26,29]
7	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}\right]$	[19,22,25,30]
8	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{1}{2}\right]$	[20,21,28,31]

Table 9: Wyckoff site: 8i, site symmetry: m2m.

No.	position	mapping
1	$[x, 0, 0]$	[1,2,7,8]
2	$[-x, 0, 0]$	[3,4,5,6]
3	$[0, x, 0]$	[9,11,14,16]
4	$[0, -x, 0]$	[10,12,13,15]
5	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[17,18,23,24]
6	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	[19,20,21,22]
7	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[25,27,30,32]
8	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	[26,28,29,31]

Table 10: Wyckoff site: `8j`, site symmetry: `m2m.`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	`[1,2,7,8]`
2	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[3,4,5,6]`
3	$\left[\frac{1}{2},\ x,\ 0\right]$	`[9,11,14,16]`
4	$\left[\frac{1}{2},\ -x,\ 0\right]$	`[10,12,13,15]`
5	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[17,18,23,24]`
6	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	`[19,20,21,22]`
7	$\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[25,27,30,32]`
8	$\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[26,28,29,31]`

Table 11: Wyckoff site: `16k`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[1,27]`
2	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[2,26]`
3	$\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[3,25]`
4	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[4,28]`
5	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[5,31]`
6	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[6,30]`
7	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[7,29]`
8	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[8,32]`
9	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	`[9,19]`
10	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	`[10,18]`
11	$\left[x+\frac{1}{2},\ x,\ \frac{3}{4}\right]$	`[11,17]`
12	$\left[\frac{1}{2}-x,\ -x,\ \frac{3}{4}\right]$	`[12,20]`
13	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	`[13,23]`
14	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	`[14,22]`
15	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{4}\right]$	`[15,21]`
16	$\left[x+\frac{1}{2},\ x,\ \frac{1}{4}\right]$	`[16,24]`

Table 12: Wyckoff site: `16l`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,8]`
2	$[x,\ -y,\ 0]$	`[2,7]`
3	$[-x,\ y,\ 0]$	`[3,6]`
4	$[-x,\ -y,\ 0]$	`[4,5]`
5	$[-y,\ x,\ 0]$	`[9,14]`
6	$[y,\ -x,\ 0]$	`[10,13]`
7	$[y,\ x,\ 0]$	`[11,16]`
8	$[-y,\ -x,\ 0]$	`[12,15]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[17,24]`
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[18,23]`
11	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[19,22]`
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[20,21]`

continued ...

Table 12

No.	position	mapping
13	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[25,30]
14	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[26,29]
15	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[27,32]
16	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[28,31]

Table 13: Wyckoff site: 16m, site symmetry: ..m'

No.	position	mapping
1	$[x,\ x,\ z]$	[1,16]
2	$[x,\ -x,\ -z]$	[2,13]
3	$[-x,\ x,\ -z]$	[3,14]
4	$[-x,\ -x,\ z]$	[4,15]
5	$[-x,\ -x,\ -z]$	[5,12]
6	$[-x,\ x,\ z]$	[6,9]
7	$[x,\ -x,\ z]$	[7,10]
8	$[x,\ x,\ -z]$	[8,11]
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[17,32]
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[18,29]
11	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[19,30]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[20,31]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[21,28]
14	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[22,25]
15	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[23,26]
16	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[24,27]

Table 14: Wyckoff site: 16n, site symmetry: .m.

No.	position	mapping
1	$[0,\ y,\ z]$	[1,6]
2	$[0,\ -y,\ -z]$	[2,5]
3	$[0,\ y,\ -z]$	[3,8]
4	$[0,\ -y,\ z]$	[4,7]
5	$[-y,\ 0,\ z]$	[9,15]
6	$[y,\ 0,\ z]$	[10,16]
7	$[y,\ 0,\ -z]$	[11,13]
8	$[-y,\ 0,\ -z]$	[12,14]
9	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[17,22]
10	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[18,21]
11	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[19,24]
12	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[20,23]
13	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[25,31]
14	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[26,32]
15	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[27,29]
16	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[28,30]

Table 15: Wyckoff site: 32o, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[x,\ -y,\ -z]$	[2]
3	$[-x,\ y,\ -z]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$[-x,\ y,\ z]$	[6]
7	$[x,\ -y,\ z]$	[7]
8	$[x,\ y,\ -z]$	[8]
9	$[-y,\ x,\ z]$	[9]
10	$[y,\ -x,\ z]$	[10]
11	$[y,\ x,\ -z]$	[11]
12	$[-y,\ -x,\ -z]$	[12]
13	$[y,\ -x,\ -z]$	[13]
14	$[-y,\ x,\ -z]$	[14]
15	$[-y,\ -x,\ z]$	[15]
16	$[y,\ x,\ z]$	[16]
17	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[17]
18	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[18]
19	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[19]
20	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[20]
21	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[21]
22	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[22]
23	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[23]
24	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[24]
25	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[25]
26	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[26]
27	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[27]
28	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[28]
29	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[29]
30	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[30]
31	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[31]
32	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[32]